

Optimization of Support Layer in Foldable OLED Module

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Abstract

With the advancement of flexible OLED display technology, foldable display modules have emerged as a crucial solution for achieving both portability and large screen display. Foldable display modules are composed of multiple heterogeneous material layers bonded by optical clear adhesive (OCA). However, due to significant differences in the mechanical properties of each layer, issues such as interlayer peeling and film fracture are prone to occur during the bending process. This study proposes an optimized design approach for the metal support layer structure. By adjusting the modulus distribution of the support layer to mitigate stress concentrations in bending regions and enhance the bending reliability of the module. Finite element analysis (FEA) was employed to simulate the bending process of the module, with particular attention to characterizing the crease defects. The results demonstrate that the optimized support layer effectively reduces stress peaks and mitigates crease formation, providing valuable insights and references for the reliability design of foldable OLED modules.

Author Keywords

Foldable OLED Module; Pattern Design; Structural Optimization; Crease.

1. Introduction

Organic Light-emitting Diode (OLED) display devices, characterized by their ultra-thin and lightweight design, bendability, and high color gamut, are widely utilized in the field of electronic displays. Foldable OLED screens have emerged as a novel form of smart terminal displays, with the foldable display market gradually expanding[1, 2]. The core component of foldable mobile terminals is the flexible OLED display module, which is composed of multiple layers stacked together to form a layered structure. These layers are bonded using Optical Clear Adhesive (OCA).

Within the OLED module, the support layer is a critical component that determines the thickness, weight, bendability, and impact resistance of the screen. Currently, support layers are typically made from metallic materials such as stainless steel, titanium alloys, and copper alloys[3]. To accommodate the bendability of OLED modules, the bending area of the support layer is often designed with a hollowed-out through-hole structure. By modifying the structural dimensions of these holes, the deformation consistency during the bending process can be adjusted. This study focuses on the mechanical behavior of the metallic support layer in flexible OLED modules during bending. Through finite element simulation, the research explores flexible design solutions for the metallic support layer structure, aiming to optimize the modulus distribution in the bending region. The goal is to reduce local stress concentrations and enhance the bending reliability of the OLED module.

2. Simulation model and optimization of metal support layer structure

2.1 Simulation model

The bending process of a flexible OLED panel was simulated using the finite element software ABAQUS, with the model simplified as a two-dimensional plane strain problem. The boundary conditions and model structure are illustrated in Figure 1. The display module has a length of 150 mm, and to ensure the flatness of the module, a thin stainless steel plate is typically used as the support layer at the bottom. Due to the symmetry of the module structure, a two-dimensional symmetric model was established for the simulation. Each film and adhesive layer was discretized into four layers of mesh, with the mesh type being linear reduced-integration plane strain elements. The film layers are composed of polymer materials and modeled using a linear elastic constitutive model. The adhesive layer exhibits both hyperelastic and viscoelastic properties, with the hyperelastic behavior described by a third-order reduced polynomial (Yeoh model) and the viscoelastic behavior represented by a Prony series in the form of a generalized Maxwell model. The material parameters and thicknesses of each layer from top to bottom are listed in Tables 1 and 2. In this study, a water-drop-shaped bending method was employed to ensure tight folding of the screen while minimizing bending marks. The foldable phone frame was modeled as a rigid body attached to the flexible panel to drive the screen's motion.

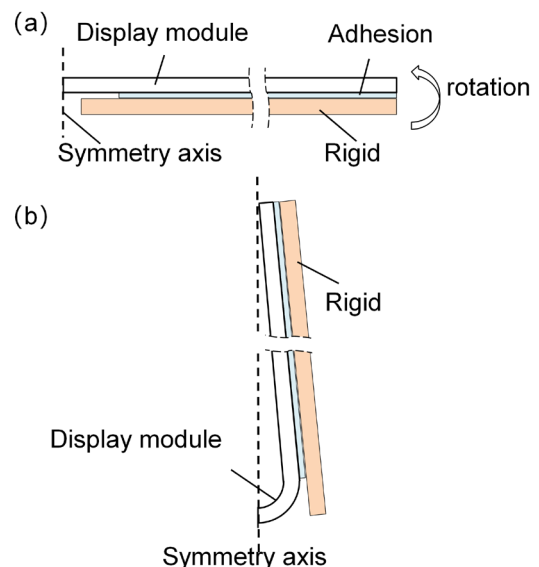


Figure 1. Schematic Diagram of the Module Bending Process. (a) Before bending. (b) After bending

Table 1. Thickness and mechanical parameters of each layer in the OLED module

Panel components	Thickness(μm)	Elastic modulus(MPa)	Poisson's ratio
CW-1	55	5500	0.33
OCA1	25	Hyperelastic & Viscoelastic	
CW-2	62	5600	0.33
OCA2	50	Hyperelastic & Viscoelastic	
Polarizer	31	3769	0.33
OCA3	15	Hyperelastic & Viscoelastic	
DOT	3	32100	0.33
TFE	14.2	25130	0.33
Array	7.56	31630	0.33
PI	16	9100	0.33
OCA4	25	Hyperelastic & Viscoelastic	
Backplane	25	3500	0.33
OCA5	25	Hyperelastic & Viscoelastic	
Foam	100	Hyperfoam	
OCA6	25	Hyperelastic & Viscoelastic	
Metal supporting	150	194000	0.33

Table 2. The hyperelastic and viscoelastic parameters of the OCA

Hyperelastic parameters			Viscoelastic parameters		
C_{10} (MPa)	4.8204E-3	g_1	0.3928	τ_1 (s)	0.0218
C_{20} (MPa)	-1.5020E-4	g_2	0.2265	τ_2 (s)	0.2878
C_{30} (MPa)	5.1208E-6	g_3	0.1446	τ_3 (s)	4.3123
D_t (MPa)	0	g_4	0.0761	τ_4 (s)	124.4600

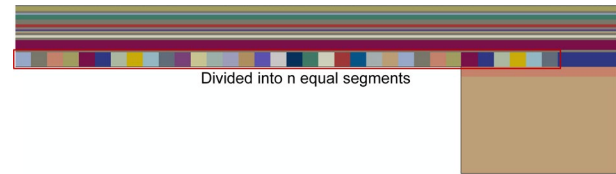
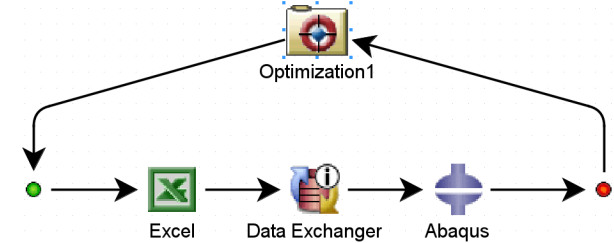
2.2 Optimization of Modulus Distribution in the Metal Support Layer

Due to the significant difference in modulus between the layers of the module, the modulus of the OCA is only a few MPa, while the modulus of the metal support layer reaches 194 GPa. The modulus difference between functional layers ranges from 100 to 1000 times. During the bending process of the module, the deformation of adjacent layers becomes inconsistent, leading to defects such as panel damage, module peeling, and crease, which severely impact the user experience[4-6]. Therefore, while ensuring the overall stiffness and supportability of the non-bending regions of the module, it is necessary to consider appropriate design modifications to the bending region of the metal support layer to reduce or avoid bending failures.

The design of patterned holes in the metal support layer is closely related to the modulus distribution. In the initial design, the patterned structure was uniformly distributed, resulting in a suboptimal stress distribution in the module. After bending, the OCA at the supported areas bears greater forces, leading to defects such as module peeling and edge bulging. In this study, the pressure borne by the OCA is taken as the optimization objective, and the Isight optimization tool is utilized to achieve a non-uniform modulus distribution in the metal support layer, thereby reducing the pressure on the OCA. The specific approach involves dividing the bending region of the initially uniformly distributed metal support layer into n equal segments, as shown in Figure 2. The modulus distribution in the bending region is adjusted to be non-uniform, assuming the modulus follows a cubic formula (Equation 1):

$$E_i = ax^3 + bx^2 + cx + d \quad (1)$$

Where E_i represents the modulus at position, and x denotes the distance from the symmetric axis. When $x = 0$, E_i is the modulus of the original pattern (3 MPa), and thus $d = -3$. The initial modulus distribution parameters are set as $a = 0.1$, $b = 1$ and $c = 10$. The Nonlinear Programming by Quadratic Lagrangian (NLPQL) algorithm is selected, and modulus optimization is performed to adjust the values of a , b , and c to minimize the pressure on the OCA. The optimization process is illustrated in Figure 3. In this study, the bending region of the metal support layer is divided into 100 equal segments. For ease of operation during simulation, a Python script is employed to batch-process tasks such as part sections and material property assignment. The modulus values of each segment are stored in an Excel table and called in Isight for optimization.

**Figure 2.** Segmentation results of the bending region in the metal support layer**Figure 3.** Flowchart of the modulus distribution optimization for the support layer

3. Results and Discussion

3.1 Module Failure Risk Assessment

After optimization using Isight, the parameters for the modulus distribution were determined as $a = 0.6062$, $b = 2.0316$ and $c = 20.0022$. By comparing the modulus differences in the bending region before and after optimization, it is observed that the modulus of the patterned SUS gradually increases from the symmetric axis to the non-bending region. This phenomenon leads to a distinct morphology of the bending region, as illustrated by the modulus distribution curve of the metal support layer in Figure 4. At the junction of the bending and non-bending regions, the modified modulus results in a smoother transition in the bending morphology of the module. Moreover, the optimized module exhibits a smaller bending radius, which effectively reduces the overall thickness of the foldable device. The morphological comparison of the module before and after optimization of the SUS support layer is presented in Figure 5. Additionally, the pressure distribution on the OCA within the module before and after optimization is depicted in Figure 6. The maximum pressure on the OCA decreases from an initial value of -0.398 MPa to -0.2203 MPa, thereby significantly reducing the risk of module peeling failure.

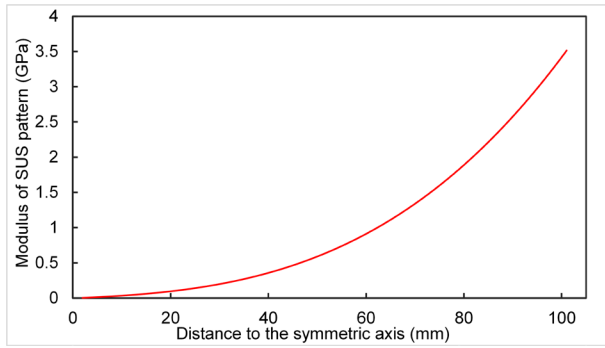


Figure 4. Optimization results of the modulus distribution for the support layer

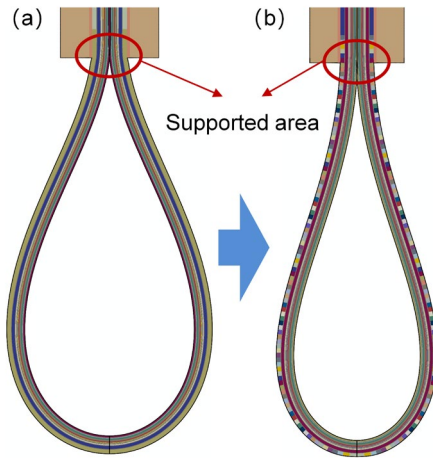


Figure 5. Comparison of module bending morphology before and after support layer modulus optimization. (a) Original structure. (b) Optimized structure.

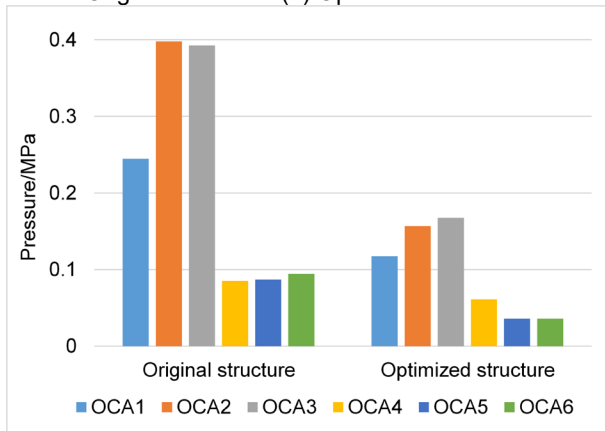


Figure 6. Comparison of pressure borne by the OCA

3.2 Analysis of Crease Degree in the Module

After optimizing the support layer structure, the module achieves a smaller bending curvature radius. However, a smaller bending radius typically increases the stress in each layer of the flexible OLED module within the bending region, exacerbating the issue of module crease. Therefore, it is necessary to evaluate the severity of crease in the module before and after optimization. In the current display industry, the crease severity of flexible display modules after flattening is generally assessed using the surface height difference and the maximum curvature. Based on the aforementioned finite element model, the module is bent and held

for 3600 seconds before being unfolded. The surface height curve of the module after flattening and resting is extracted. The highest and lowest points are identified from the surface height curve, and the difference between them is defined as the surface height difference. However, when the surface height difference is the same, a steeper curve slope indicates more pronounced crease. Therefore, the maximum surface curvature is introduced. The curvature k is calculated using Equation (2):

$$k = 1 / R = 2 \sin A / a \quad (2)$$

where a represents the side length of a triangle formed by three adjacent points on the surface curve, and R is the circumradius of the triangle.

The surface curve and corresponding surface curvature of the flexible OLED display module after flattening and resting are shown in Figure 7. The surface height difference and the maximum curvature of the optimized module are significantly reduced, indicating that the non-uniform modulus distribution of the support layer effectively improves the crease severity of the module.

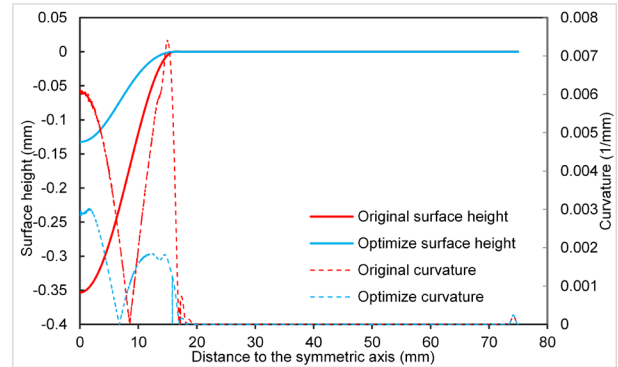


Figure 7. Surface height curve and corresponding curvature of the module

4. Conclusions

This paper proposes an optimization scheme for the non-uniform modulus distribution of the metal support layer. The optimized module exhibits a smoother transition in the water-drop-shaped bending morphology, which not only diminishes the risk of stress concentration but also enhances deformation compatibility between the support layer and adjacent layers. This approach effectively mitigates the risk of module peeling failure, concurrently reducing the bending radius and alleviating the severity of module creasing. The proposed method provides valuable insights for the reliability design of flexible OLED modules.

5. Acknowledgements

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